

Financial Econometrics

Panel Data Analysis

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Does managerial ownership affect firm value?

As we already discussed in the Lab Session of Week 6 - Non-linear functional form, managerial ownership might affect firm value. Please see the corresponding Lab session, for the economic rationale. In this lab session, we investigate if we find stronger or weaker support of a positive effect if exploiting a panel set-up.

Please use *MB_Own.Rdata*, a data-frame containing following variables:

- *YEAR* is fiscal year
- *GVKEY* is company identifier
- *mtbe* is market-to-book
- *own* is CEO ownership

Task 1: Prepare the data

- Load the data
- Remove missing values using *na.omit()*
- Remove erroneous entries (e.g., negative *mtbe*)
- Winsorize *own* and *mtb* at the 99% level
- Declare it to be a panel data. Please see the coding hint and see *?pdata.frame*:

```
# package
require(plm)

# declare data-frame as pdata-frame
... <- pdata.frame(..., index = c("...", "..."))
```

Task 2: Equivalence of “first difference”, “binary regressors”, and “entity-demeaned” approach using only the years 2000 and 2008

- Limit the data to the years $t = 2000$ and $t = 2008$ and firms i with complete data in both years
- Estimate following benchmark model

$$mtbe_{t,i} = \alpha + \beta_1 \times own_{t,i} + u_{t,i}$$

- Estimate following model that has firm fixed effects α_i

$$mtbe_{t,i} = \alpha_i + \beta_1 \times own_{t,i} + u_{t,i}$$

using three different estimation techniques:

- First difference approach: *fit = plm(y~0 + x, data = ..., model = "fd")*. Please note the “0” indicating that the constant will be dropped. Otherwise the coefficients would not be identical.

- Binary regressor: $fit = plm(y \sim x + factor(firm\ identifier), data = \dots, model = "pooling")$
- Entity-demeaned approach: $fit = plm(y \sim x, data = \dots, model = "within", effect = "individual")$.
- Please cluster standard errors by firm for all regressions, except for model 3, “binary regressors”. While you should always cluster standard errors, you might run out of memory for model 3, hence drop it for now.

```
require(car)
require(lmtest)
require(sandwich)

# clustered standard errors
coeftest(fit, vcov = vcovHC(fit, cluster = "group", type = "sss"))
```

- Make a stargazer table summarising the output. Use `stargazer(..., omit = "GVKEY")` to avoid that the table contains all the fixed effects from the binary regressor model.
- Does it matter if we include firm fixed effects or not? Why?

Task 3: Full sample estimation

Please estimate following models using the full sample:

- Estimate following benchmark model

$$mtbe_{t,i} = \alpha + \beta_1 \times own_{t,i} + u_{t,i}$$

- Estimate following model that has time fixed effects λ_t

$$mtbe_{t,i} = \lambda_t + \beta_1 \times own_{t,i} + u_{t,i}$$

- using a time-demeaned approach: $fit = plm(y \sim x, data = \dots, model = "within", effect = "time")$.

- Estimate following model that has firm fixed effects α_i and time fixed effects λ_t

$$mtbe_{t,i} = \alpha_i + \lambda_t + \beta_1 \times own_{t,i} + u_{t,i}$$

using two different estimation techniques

- Entity-demeaned approach with explicit year fixed effects: $fit = plm(y \sim x + factor(YEAR), data = \dots, model = "within", effect = "individual")$.
- Entity-demeaned and time-demeaned approach: $fit = plm(y \sim x, data = \dots, model = "within", effect = "twoway")$.
- Make a stargazer table summarising the output. Use `stargazer(..., omit = "YEAR")` to avoid that the table contains all the fixed effects from the Binary Regressor model.
- Does it matter if we include firm fixed effects or not? Why?

```

# Solution

# packages
require(car)
require(lmtest)
require(dplyr)
library(sandwich)
require(DescTools)
require(stargazer)
require(plm)

# Clear enviroment
rm(list=ls(all=TRUE))

# working directory

setwd("C:/Users/s13163/Dropbox/FIE401/data/data_labs")

# Task 1: Prepare data

# load
load("MB_Own.Rdata")

# change name
own <- MB_Own
rm(MB_Own)

# summary stat
summary(own)

##          GVKEY          YEAR          mtbe          own
## Min.   : 1004   Min.   :1995   Min.   :0.01354   Min.   :0.00000
## 1st Qu.: 7065   1st Qu.:2000   1st Qu.:1.45145   1st Qu.:0.00104
## Median : 13501   Median :2006   Median :2.20020   Median :0.00355
## Mean   : 37594   Mean   :2005   Mean    : Inf     Mean   :0.02300
## 3rd Qu.: 31223   3rd Qu.:2010   3rd Qu.:3.59146   3rd Qu.:0.01316
## Max.   :316056   Max.   :2015   Max.    : Inf     Max.   :1.29547
##                                     NA's    :918

# omit missing observations
own <- na.omit(own)

# Drop observations with negative mtbe
# Be attentive there are no negative mtbe observations!!!

# Drop observation with own>100
own <- own[(own$own<=1),];

# Drop observation with mtbe=inf
own<-own[(own$mtbe!=Inf),];

# summary stats updated
summary(own)

##          GVKEY          YEAR          mtbe          own
## Min.   : 1004   Min.   :1995   Min.   : 0.0135   Min.   :0.000000
## 1st Qu.: 7070   1st Qu.:2000   1st Qu.: 1.4512   1st Qu.:0.001038
## Median : 13561   Median :2006   Median : 2.2004   Median :0.003553
## Mean   : 37791   Mean   :2005   Mean    : 4.5843   Mean   :0.022961
## 3rd Qu.: 31453   3rd Qu.:2011   3rd Qu.: 3.5990   3rd Qu.:0.013155
## Max.   :316056   Max.   :2015   Max.    :8050.0000   Max.   :0.811335

# winsorize
own$own <- Winsorize(own$own, val=quantile(own$own,probs = c(0.005, 0.995), na.rm=TRUE))
own$mtbe <- Winsorize(own$mtbe, val=quantile(own$mtbe,probs = c(0.005, 0.995), na.rm=TRUE))

# Task 2: Equivalence of approaches

```

```

# extract firms that appear in year 2000 and 2008
filter(own, YEAR == 2000) %>% pull(GVKEY) -> gvkey.2000
filter(own, YEAR == 2008) %>% pull(GVKEY) -> gvkey.2008

# intersect
gvkey.selected <- intersect(gvkey.2000, gvkey.2008)

# limit the data
own %>%
  # limit based on years
  filter(YEAR %in% c(2000, 2008)) %>%
  # limit based on selected firms
  filter(GVKEY %in% gvkey.selected) -> own.2

# declare panel data frame
own.2 <- pdata.frame(own.2, index=c("GVKEY", "YEAR"))

# check
pdim(own.2)

## Balanced Panel: n = 1007, T = 2, N = 2014

# model 1: benchmark
fit1 <- plm(mtbe ~ own, data = own.2, model = "pooling")
robust_se1 <- coeftest(fit1, vcov=vcovHC(fit1, cluster="group", type="sss"))[,2]

# model 2: First differencing
fit2 <- plm(mtbe ~ 0 + own, data = own.2, model="fd")
robust_se2 <- coeftest(fit2,vcov=vcovHC(fit2, cluster="group", type="sss"))[,2]

# Binary regressors
fit3 <- plm(mtbe ~ own + factor(GVKEY), data = own.2, model="pooling")
# robust_se3 <- coeftest(fit3,vcov=vcovHC(fit3, cluster="group", type="sss"))[,2]
# Note that robust_se3 might be not computed as you are likely to run out of memory

# Entity-demeaned approach
fit4 <-plm(mtbe ~ own, data = own.2, model="within", effect="individual")
robust_se4 <- coeftest(fit4,vcov=vcovHC(fit4, cluster="group", type="sss"))[,2]

# output
stargazer(list(fit1, fit2, fit3, fit4),
  se=list(robust_se1, robust_se2, NULL, robust_se4),
  type="text",
  keep.stat=c("n", "rsq", "adj.rsq"),
  omit = "GVKEY",
  report=('vc*t'))

##
## =====
##                               Dependent variable:
##                               -----
##                               mtbe
##                               (1)      (2)      (3)      (4)
## -----
## own                3.375**   9.307***  9.307***  9.307***
##                   t = 1.979   t = 4.937  t = 3.316  t = 4.937
##
## Constant           2.828***                0.881
##                   t = 29.330                t = 0.353
##
## -----
## Observations       2,014        1,007        2,014        2,014
## R2                  0.002        0.001        0.584        0.011
## Adjusted R2        0.002        0.001        0.167       -0.979
## =====
## Note:                  *p<0.1; **p<0.05; ***p<0.01

```

```

# Task 3: full data

# declare panel data frame
own <- pdata.frame(own, index=c("GVKEY","YEAR"))

# check
pdim(own)

## Unbalanced Panel: n = 3467, T = 1-21, N = 35301

# model 1: benchmark
fit1 <- plm(mtbe ~ own, data = own, model = "pooling")
robust_se1 <- coeftest(fit1, vcov=vcovHC(fit1,cluster="group", type="sss"))[,2]

# Model 2: time fixed effects
fit2 <- plm(mtbe ~ own , data = own, model="within", effect = "time")
robust_se2 <- coeftest(fit2,vcov=vcovHC(fit2, cluster="group", type="sss"))[,2]

# Model 3: time and firm fixed effects
fit3 <- plm(mtbe ~ own + factor(YEAR), data = own, model="within", effect = "individual")
robust_se3 <- coeftest(fit3,vcov=vcovHC(fit3, cluster="group", type="sss"))[,2]

# Model 4: time and firm fixed effects
fit4 <-plm(mtbe ~ own , data = own, model="within", effect="twoway")
robust_se4 <- coeftest(fit4,vcov=vcovHC(fit4, cluster="group", type="sss"))[,2]

# output
stargazer(list(fit1, fit2, fit3, fit4),
           se=list(robust_se1, robust_se2, robust_se3, robust_se4),
           type="text",
           keep.stat=c("n","rsq","adj.rsq"),
           omit = "YEAR",
           report=('vc*t'))

```

```

##
## =====
##                               Dependent variable:
##                               -----
##                               mtbe
##                               (1)      (2)      (3)      (4)
## -----
## own                2.525**    2.215**  4.429***  4.429***
##                   t = 2.330  t = 2.058 t = 4.289 t = 4.290
##
## Constant           3.300***
##                   t = 59.152
##
## -----
## Observations      35,301    35,301    35,301    35,301
## R2                 0.001     0.001     0.030     0.002
## Adjusted R2       0.001     0.0002    -0.076    -0.108
## =====
## Note:                *p<0.1; **p<0.05; ***p<0.01

```